

Instructor: Date:

Learner: Regional Branch:

Program: Center:

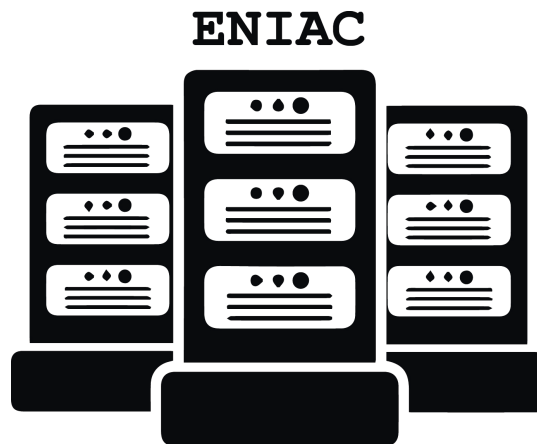
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Let's Start!

Let's go back to the early years of computers.



Learning Outcome:

Identify information about the history of software and its characteristics in simple spoken and written situations.
Give information about software history, characteristics and methodologies.

Can Do Statements:

- I can talk about characteristics of the first electronic modern computers.
- I can say what the Von Neumann architecture is.
- I can talk about the role of software in the first electronic modern computers.

Ask your partner:

- How do you imagine the first computers? Try to describe them.
- Why was the invention of the computer an important achievement for humankind?





Let's Explore!

Gale talks about electronic computers.



Instructions: Read the text below.

Gale: I am ready for my test.

Teacher: Ok. What were the characteristics of the first electronic computers?

Gale: Well, the first programmable computer was the ENIAC (Electronic Numerator Integrator Analyzer and Computer). In 1942, John W. Mauchly made a summary of the idea of a programmable computer and sent it to the government of the United States. After some time, in June 1943, the US military listened to Mauchly's idea and was ready to start the construction of this modern computer.

Teacher: Describe the main functions of the ENIAC at that time.

Gale: The machine had military purposes. Umm, let's suppose the machine is right in front of us. The first thing that we can notice is the enormous size; that's why many operators were necessary to use it. The machine uses vacuum tubes and it helps calculate ballistics tables.

Teacher: You're right. Those tables were important for soldiers to use their weapons correctly. What about EDVAC?

Gale: It stands for Electronic Discrete Variable Computer. John von Neumann's ideas were important for creating this machine.

Teacher: Who was John von Neumann?

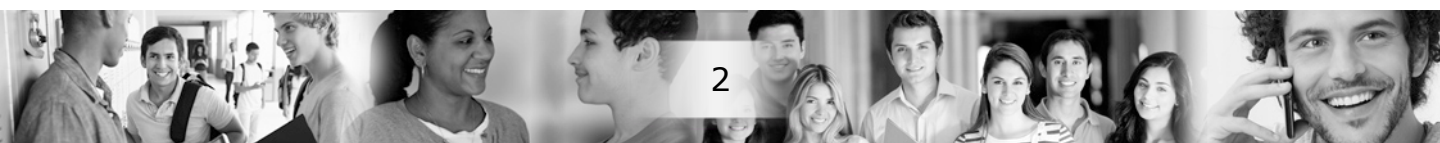
Gale: He was the creator of the Von Neumann architecture. In 1945, he and his coworkers wrote about this architecture in the EDVAC report. The von Neumann architecture describes the functioning of the programmable computer. The key components of the architecture are: arithmetical unit, control unit, memory, input, and output.

Teacher: Now, tell me about the UNIVAC.

Gale: The Universal Automatic Computer was the first commercial computer. The United States Census Bureau first used it on March 31, 1951. Computers like the ENIAC, EDVAC, and UNIVAC were very big and expensive machines.

Teacher: Were those machines the first computers?

Gale: Yes. Now, all those machines used software, but, at that time, the role of software in the first electronic computers was incipient because programmers used machine language or programmed computers mechanically.





Let's Practice!

Defining John Von Neumann's* Architecture for Electronic Computers.

*"John von Neumann" was a Hungarian mathematician. He was born on December 28, 1903.

Instructions: Match the sentences from Column A with Column B.

COLUMN A

1. It is a computer architecture.
Arithmetic unit, control unit, memory,
input and output...
2. The development of the ENIAC and
the Neumann Architecture were...
3. The Turing Machine and the Neumann
Architecture were...

COLUMN B

- a. ... different because the Neumann
Architecture used a RAM memory.
- b. ... were the five main components of
a programmable computer proposed
by von Neumann.
- c. ... very important for the creation of
the computers we know today.



Let's Practice!

The First Programmable* Machines.

*"Programmable" refers to any device that can interpret a code.

Instructions: Select the correct answer to complete the sentence.

1. ____ a specific time during the World War II, the first electronic computer appeared.
2. The first commercial automatic computer was first used ____ March 31, 1951. It was called UNIVAC.
3. ____ June 1943, the construction of the ENIAC started.





Extension Activity

It's vital to organize information chronologically.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for further information about the early years of programmable machines and software.
- Identify the important dates in the history of programmable computers.
- Create a timeline using the previous information. Include pictures and key sentences.
- Share your work with a classmate.

Self-Assessment

Yes

No

- I look for relevant information about the early years of programmable machines and software.
- I identify the dates that are essential in the history of programmable computers and organize them chronologically.
- I create a timeline and share it with a classmate.

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Target Words

- **bureau** (verb): a government department
e.g. The United States Census Bureau first used it on March 31, 1951.
- **ballistics** (noun): a study of the motion of projectiles
e.g. The machine uses vacuum tubes and it helps calculate ballistics tables.
- **incipient** (adjective): beginning to exist
e.g. The role of software in the first electronic computers was incipient because programmers used machine language or programmed computers mechanically.
- **vacuum tube** (noun): a glass tube that contains electrodes
e.g. The machine uses vacuum tubes and it helps calculate ballistics tables.
- **notice** (verb): to realize about something
e.g. Let's suppose the machine is right in front of us. The first thing that we can notice is the enormous size of the computer.
- **weapon** (noun): a gun or another artefact that you use to attack someone or defend yourself
e.g. Those tables were important for soldiers to use their weapons correctly.



Learning Points

Simple Past Tense of Verb Be, Were in Affirmative Statements

The verb to be in past takes two different forms, which are **was** and **were**. The use of those two forms depends on the pronoun you use, let's see how it works.

Take a look at the following chart:

Pronoun	Verb to be
I	was
You (singular and plural)	were
He / She	was
We	were
They	were
It	was

As you can see, the conjugation of verb to be in past depends on the pronoun. Now, when you want to use **were** in an affirmative sentence, you have to follow the structure below:

Subject + **verb to be in past** (in this case: **were**) +
complement (it is commonly an adjective).

Examples:

*The students **were** very curious about the development of the first electronic computers.*
*The first computers **were** too big.*

Examples

- *John von Neumann's ideas were important for creating this machine.*
- *The soldiers were ready to use the ENIAC.*
- *Computers like the ENIAC, EDVAC and UNIVAC were very big and expensive machines.*





Learning Points

Prepositions of Time

Prepositions are words that you use before or after nouns or verbs in order to give information about a place or a time. In this case, we're going to focus on **prepositions of time**.

In the following chart, you are going to find a description of each **preposition of time**:

Preposition	Uses
At	- We use it with clock times and meal times: <i>My software class is at 09:30.</i> <i>We always share our ideas at lunchtime.</i> - We use it to talk about holidays: <i>We celebrate at Christmas.</i> - In certain fixed expressions which refer to specific moments: <i>...at the moment.</i> <i>...at the end of the year.</i> <i>...at the same time.</i> <i>...at that time.</i>
On	- You use it for talking about days and dates: <i>He's going to talk about the first electronic computers on Monday.*</i> <i>Our exam about software is on the 10th of July.*</i> *Remember that in English language you have to use capital letters when writing the days of the week and the months of the year.
In	- You use it to talk about months, years decades and centuries: <i>The ENIAC was officially released in 1946.</i> <i>To be more precise, it was released in February 1946.</i> <i>The first software developments were made in the 19th century.</i> <i>In the 1950s, the ENIAC was deactivated.</i> - You can also use it with seasons and moments of the day: <i>...in summer.</i> <i>...in the evening.</i> <i>...in the morning.</i>



Learning Points

Examples

- In 1942, John W. Mauchly made a summary of the idea of a programmable computer and sent it to the government of the United States.
- The United States Census Bureau first used the UNIVAC on March 31, 1951.
- After some time, in June 1943, the US military listened to Mauchly's idea and was ready to start the construction of this modern computer.
- Describe the main functions of the ENIAC at that time.

The von Neumann Architecture

The creation of the Electronic, Numerator, Integrator, Analyzer and Computer (ENIAC) in 1942 facilitated different developments in the field of programmable computers. The Electronic Discrete Variable Computer (EDVAC), created in 1945, was one of such developments. This machine included the **von Neumann Architecture**.

The von Neumann architecture describes a design architecture for electronic or programmable computers. This architecture is still famous nowadays and it has five key components: arithmetical unit, a control unit, memory, input and output.

Let's describe the five components of the architecture:

Component	Description
Arithmetic Logic Unit (ALU)	Think of ALU as an electronic unit that contains different mathematical operations.
Control Unit	This part of the architecture is in charge of moving data and programming instructions. You can describe it as the component that guarantees the quality of the computer's performance.
Memory	The memory is in charge of saving all the information in the programmable computer.
Input / Output	On one hand, input refers to the information that you send to the computer. On the other hand, output is related to the information that the computer processes. Nowadays, an input device can be a monitor or a keyboard and an output device can be a printer.